Airline Operational Audit

Analyizing Flight Delay Trends for *Frontier Airlines*

This analysis focuses on carrier delay occurrences and delay times for Frontier airlines. We break this analysis into four sections each with their own summary figures and interpretations. The libraries used were tidyverse, ggthemes, gridextra, lubridate, gt, and scales. The main alterations we had to make to the data were to filter it for our airline and to fill NA values. If a flight experienced no delays, the dataset would include it as NA instead of zero, so we had to account for that.

Analysis of Airports with the Most Delays

In this first section, we looked at the worst performing airports that Frontier services by carrier delay on Frontier flights. We found both the average delay for flights out of the airport as well as proportion of flights delayed. We had to filter the airports to ensure that at least 10 flights left the airport from Frontier in that month to ensure that outliers would not skew the data. We found that the airports with the worst delay times were Pensacola, Austin-Bergstrom, San Juan, Bradley airport in Hartford, Ct, and Gerald R. Ford airport in Grand Rapids, MI. The airports with the highest chance of a delayed flight were Liberty Airport in Newark, NJ, San Juan, Logan Airport in Boston, MA, Philadelphia, and Hopkings airport in Cleveland, OH. All of these airports would be worth further research, but San Juan would be especially important to address because it is in the top 5 for both delay times and chance of delay.

```
#Invoking Libraries
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────────
tidyverse 2.0.0 ──
✔ dplyr     1.1.4     ✔ readr     2.1.5
✔ forcats   1.0.0     ✔ stringr   1.5.2
✔ ggplot2   4.0.0     ✔ tibble    3.3.0
```

```
✔ lubridate 1.9.4     ✔ tidyr     1.3.1
✔ purrr     1.1.0
── Conflicts ────────────────────────────────────────
tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>)
to force all conflicts to become errors
```

```r
library(ggthemes)
library(gridExtra)
```

```
Attaching package: 'gridExtra'

The following object is masked from 'package:dplyr':

    combine
```

```r
data <- read_csv("2025-01.csv")
```

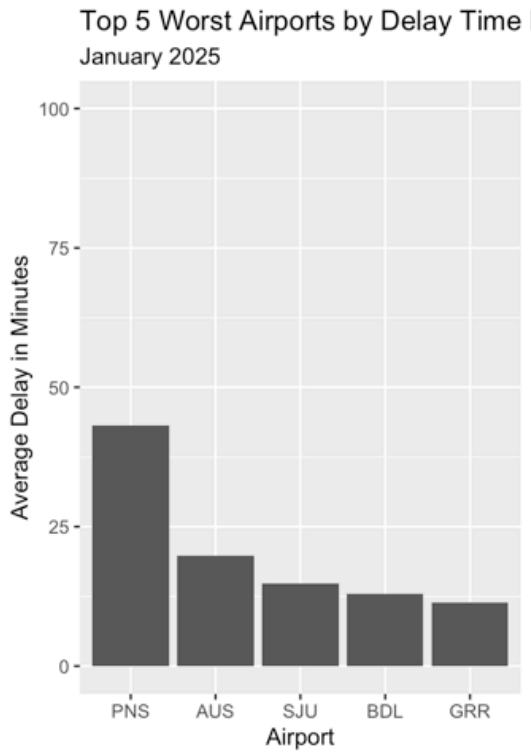
```
Rows: 599013 Columns: 26
── Column specification
────────────────────────────────────────────────────────
Delimiter: ","
chr  (9): FL_DATE, MKT_UNIQUE_CARRIER, MKT_CARRIER, ORIGIN,
ORIGIN_CITY_NAME...
dbl (17): MONTH, MKT_CARRIER_FL_NUM, ORIGIN_AIRPORT_ID,
ORIGIN_AIRPORT_SEQ_I...

ℹ Use `spec()` to retrieve the full column specification for
this data.
ℹ Specify the column types or set `show_col_types = FALSE` to
quiet this message.
```

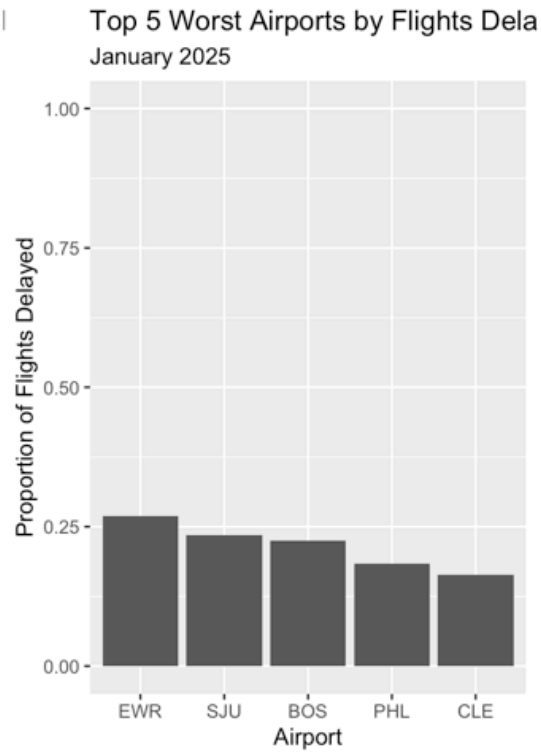
```r
data <- data %>%
  select(ORIGIN,MKT_UNIQUE_CARRIER,CARRIER_DELAY) %>%
  mutate(across(everything(),~replace_na(.,0)))

frontier_data <- data %>% filter(MKT_UNIQUE_CARRIER == 'F9')

Delay_Airport<- frontier_data %>%
  group_by(ORIGIN)%>%
  filter(n()>=15)%>%
  summarize(Average_Delay = mean(CARRIER_DELAY), Delay_Prop = su
```

```
glimpse(Delay_Airport)
```

```
Rows: 64
Columns: 3
$ ORIGIN        <chr> "ATL", "AUS", "BDL", "BNA", "BOS", "BQN",
"BTV", "BUF", …
$ Average_Delay <dbl> 8.99227373, 19.77631579, 13.01612903,
6.13157895, 7.4775…
$ Delay_Prop    <dbl> 0.15673289, 0.06578947, 0.09677419,
0.07894737, 0.224719…
```

```
Delay_Airport$ORIGIN<- factor(Delay_Airport$ORIGIN, levels = De

Prop_Delay_Airport <- Delay_Airport%>%
  arrange(desc(Delay_Prop))

Average_Delay_Airport <- Delay_Airport%>%
  arrange(desc(Average_Delay))

Average_Delay_Airport_Top <- Average_Delay_Airport %>%
  slice(1:5)

Prop_Delay_Airport_Top <- Prop_Delay_Airport%>%
  slice(1:5)

Average_Plot<- ggplot(data = Average_Delay_Airport_Top, aes(x =
  geom_bar(stat = 'identity')+
  labs(title = 'Top 5 Worst Airports by Delay Time For Frontier
       subtitle = 'January 2025',
       x = 'Airport',
       y = 'Average Delay in Minutes') + ylim(0,100)

Prop_Plot<- ggplot(data = Prop_Delay_Airport_Top, aes(x = reord
  geom_bar(stat = 'identity')+
  labs(title = 'Top 5 Worst Airports by Flights Delayed For Fro
       subtitle = 'January 2025',
       x = 'Airport',
       y = 'Proportion of Flights Delayed')+
  ylim(0,1)

grid.arrange(Average_Plot, Prop_Plot, ncol = 2)
```





Comparison of Average Delays b/w Frontier and WN

In this section we compare Frontier to the best performing airline by average delay time in minutes. The best performing airline in this case was Southwest which had far more flights as well as an average delay that was less than half of Frontier's. Southwest is a larger airline with more access to resources and routes, so some difference is inevitable, but around a two to two and a half minute average delay across all flights is somethign that Frontier should be striving for.

```
library(tidyverse)
library(gt)

data <- readr::read_csv("2025-01.csv")
```

```
Rows: 599013 Columns: 26
── Column specification
────────────────────────────────────────────────────────
Delimiter: ","
chr  (9): FL_DATE, MKT_UNIQUE_CARRIER, MKT_CARRIER, ORIGIN,
ORIGIN_CITY_NAME...
dbl (17): MONTH, MKT_CARRIER_FL_NUM, ORIGIN_AIRPORT_ID,
ORIGIN_AIRPORT_SEQ_I...
```

```
ℹ Use `spec()` to retrieve the full column specification for
this data.
ℹ Specify the column types or set `show_col_types = FALSE` to
quiet this message.
```

```
data<- data%>%
mutate(across(everything(),~replace_na(.,0)))

AirlineCode <- "F9"  # Frontier

##  treat NA as 0
avg_carrier <- data %>%
  group_by(MKT_CARRIER) %>%
  summarise(Avg_Carrier_Delay = mean(CARRIER_DELAY),
            Flights = n(),.groups = "drop") %>%
  arrange(Avg_Carrier_Delay)

#frontier vs best airline
frontier_row <- avg_carrier %>% filter(MKT_CARRIER == AirlineCo
best_row     <- avg_carrier %>% slice(1)


#Comparison table
tibble(
  Airline = c("Frontier (F9)", paste0(best_row$MKT_CARRIER, " (
  `Avg Carrier-caused Delay (min)` = c(frontier_row$Avg_Carrier_
                                       best_row$Avg_Carrier_Del
  Flights = c(frontier_row$Flights, best_row$Flights)) %>%
  mutate(`Gap vs Best (min)` = `Avg Carrier-caused Delay (min)`
  gt() %>%
  tab_header(title = md("**Frontier vs Best Airline**"),
             subtitle = md("**Metric: CARRIER_DELAY (airline-co
  cols_align("center") %>%
  fmt_number(columns = c(`Avg Carrier-caused Delay (min)`, `Gap
```

Frontier vs Best Airline

Metric: CARRIER_DELAY (airline-controlled minutes)

Airline	Avg Carrier-caused Delay (min)	Flights	Gap vs Best (min)
Frontier (F9)	5.68	15526	3.40
WN (Best)	2.28	105307	0.00

Comparison of Average Delays

This graph breaks down all the flights over January 2025 into delaye and not delayed. The worst day was January 6 when almost a quart of flights were delayed. The best day was January 28th where less than a percent of flights were delayed. It is not clear if there are any monthly trends to delays, but Frontier cold look to reduce the maximum percent of flights delayed on any day during the period.

```
library(tidyverse)
library(tidyr)
library(lubridate)
library(scales)
```

```
Attaching package: 'scales'

The following object is masked from 'package:purrr':

    discard

The following object is masked from 'package:readr':

    col_factor
```

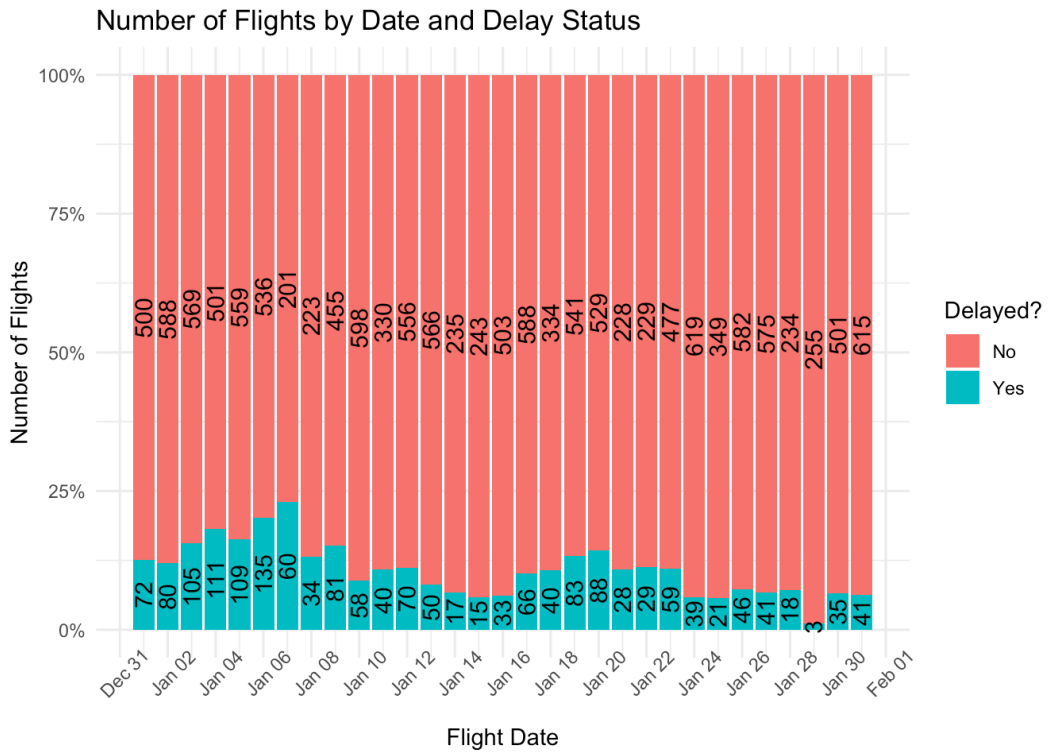
```
rm(list=ls())

data <- read_csv("2025-01.csv")
```

```
Rows: 599013 Columns: 26

── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr  (9): FL_DATE, MKT_UNIQUE_CARRIER, MKT_CARRIER, ORIGIN,
ORIGIN_CITY_NAME...
dbl (17): MONTH, MKT_CARRIER_FL_NUM, ORIGIN_AIRPORT_ID,
ORIGIN_AIRPORT_SEQ_I...

ℹ Use `spec()` to retrieve the full column specification for
this data.
ℹ Specify the column types or set `show_col_types = FALSE` to
quiet this message.
```

```
#cleaning the dataset
frontier_data <- as.data.frame(data) %>%
     filter(MKT_UNIQUE_CARRIER == "F9") %>%
     select(FL_DATE, CARRIER_DELAY) %>%
     mutate(across(everything(),~replace_na(.,0)))%>%
     mutate(delayq = case_when(CARRIER_DELAY > 0 ~ "Yes",TRUE ~

#transforming the data so it works with the group_by
frontier_data <- frontier_data %>%
  mutate(FL_DATE = parse_date_time(FL_DATE, orders = c("mdy HMS'
  drop_na(FL_DATE)

#Grouping the data before plotting
summary_df <- frontier_data %>%
     group_by(FL_DATE, delayq) %>%
     summarise(count = n()) %>%
     mutate(prop = count/sum(count))
```

```
`summarise()` has grouped output by 'FL_DATE'. You can override
using the
`.groups` argument.
```

```
#Plotting the data
ggplot(summary_df, aes(x = FL_DATE, y = prop, fill = delayq)) +
     geom_bar(stat = "identity") +
     labs(
          title = "Number of Flights by Date and Delay Status",
          x = "Flight Date",
          y = "Number of Flights",
          fill = "Delayed?") +
     theme_minimal() +
     scale_y_continuous(labels = percent) +
     geom_text(aes(label = count),
               position = position_stack(vjust = 0.5),
               color = "black", size = 4,angle=90) +
    scale_x_date(date_breaks = "2 days", date_labels = "%b %d")
     theme(axis.text.x = element_text(angle=45))
```

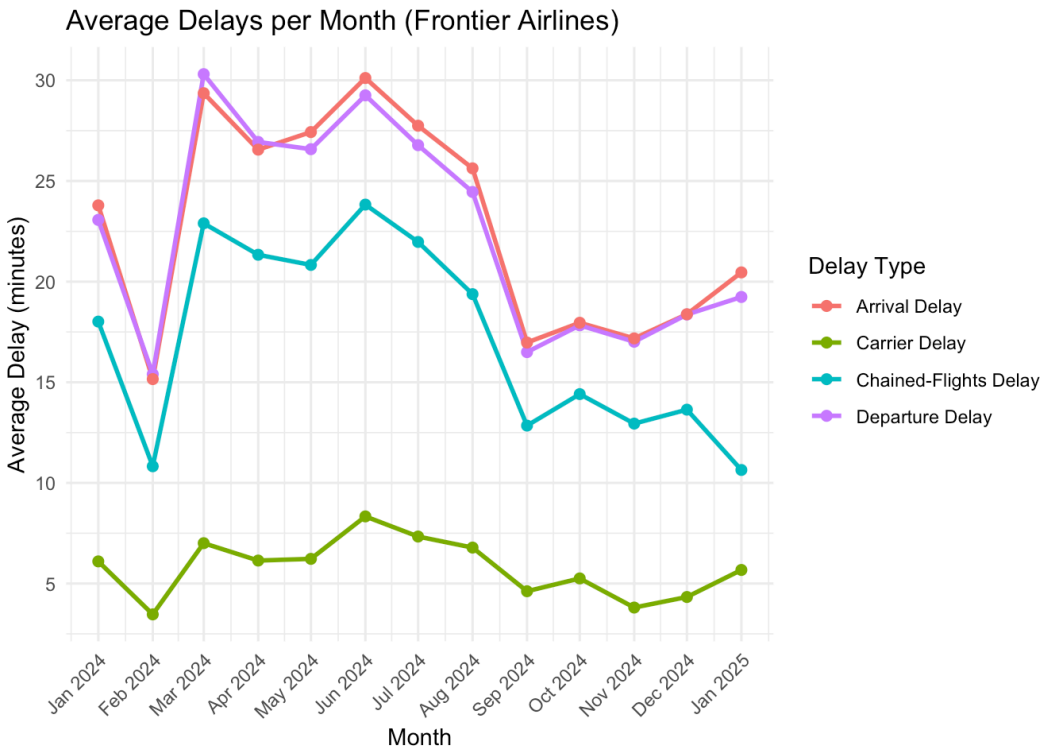


Average Delay Comparison between Departure, Arrival and Carrier Delays

So far in this analysis, we have only looked at carrier delay since it is the delay that Frontier has the most control over. However, Carrier delay does not capture the effects of chain delays, as in a delay at one airport causes the plane to arrive late at another which delays another flight. This extra delay can be captured by adding carrier delay and late aircraft delay to get a variable we call chained-delay. Other factors include weather delays and discrepancies in expected flight time which manifest in total departure delay and arrival delay. This graph plots the average of all four delays for Frontier over 2024. As expected, carrier delay is the lowest and is fairly consistent over the year. Chained delay and total departure delay tracked relatively similarly except between December 2024 and January 2025 which could be due to weather delats. Arrival delays very closely tracked departure delays with some noise due to the inherent randomness in flight times. The worst period for delay times was from March to July. Interestingly, base carrier delays were not affected much during this period, indicating that the spike could be due to the effect of delays being magnified over several consecutive flights. This indicates that Frontier may need to better manage aircraft availability within that period.

```
rm(list=ls())
```

```r
library(tidyverse)
library(lubridate)

files <- c("2024-01","2024-02","2024-03","2024-04","2024-05","2(
full <- tibble()

for (i in files){
  df <- read_csv(paste0(i,".csv")) %>%
    select(FL_DATE,MONTH,MKT_UNIQUE_CARRIER,DEP_DELAY_NEW,ARR_DI
    filter(MKT_UNIQUE_CARRIER=="F9") %>%
    mutate(across(where(is.numeric), ~replace_na(.,0))) %>%
    mutate(CARRIER_DELAY2 = CARRIER_DELAY+LATE_AIRCRAFT_DELAY) %
    mutate(MONTH = factor(MONTH)) %>%
    mutate(MKT_UNIQUE_CARRIER = factor(MKT_UNIQUE_CARRIER)) %>%
    mutate(FL_DATE = sub(" .*", "", FL_DATE)) %>%
    mutate(FL_DATE = mdy(FL_DATE), MONTH = month(FL_DATE), YEAR
  full <- bind_rows(full,df)
}
# class(full$FL_DATE)
# summary(full)
# glimpse(full)

avg_delay <- full %>%
  group_by(YEAR, MONTH) %>%
  summarise(avg_carrier = mean(CARRIER_DELAY, na.rm = TRUE),
            avg_dep = mean(DEP_DELAY_NEW, na.rm = TRUE),
            avg_arr = mean(ARR_DELAY_NEW, na.rm = TRUE),
            avg_late = mean(CARRIER_DELAY2, na.rm = TRUE),.grou
  # reshape wide → long from GPT
  pivot_longer(cols = starts_with("avg_"),
               names_to = "delay_type",
               values_to = "avg_delay") %>%
  mutate(date = as.Date(paste(YEAR, MONTH, "01", sep = "-")),
         delay_type = recode(delay_type,avg_carrier = "Carrier D
                             avg_dep = "Departure Delay",
                             avg_arr = "Arrival Delay",
                             avg_late="Chained-Flights Delay"))

ggplot(avg_delay, aes(x = date, y = avg_delay, color = delay_typ
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  scale_x_date(date_labels = "%b %Y", date_breaks = "1 month") +
  labs(title = "Average Delays per Month (Frontier Airlines)",x
       color = "Delay Type") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  scale_y_continuous(breaks = seq(0, 120, by = 5))
```


Average Delays per Month (Frontier Airlines)
Average Delay (minutes)
30
25
20
15
10
5
Jan 2024
Feb 2024
Mar 2024
Apr 2024
May 2024
Jun 2024
Jul 2024
Aug 2024
Sep 2024
Oct 2024
Nov 2024
Dec 2024
Jan 2025
Month
Delay Type
Arrival Delay
Carrier Delay
Chained-Flights Delay
Departure Delay